Министерство науки и высшего образования Российской Федерации
ФЕДЕРАЛЬНОЕ ГОСУДАРСТВЕННОЕ АВТОНОМНОЕ ОБРАЗОВАТЕЛЬНОЕ
УЧРЕЖДЕНИЕ ВЫСШЕГО ОБРАЗОВАНИЯ
НАЦИОНАЛЬНЫЙ ИССЛЕДОВАТЕЛЬСКИЙ УНИВЕРСИТЕТ ИТМО
(НИУ ИТМО)

Факультет программной инженерии и компьютерной техники

ОТЧЁТ
О ЛАБОРАТОРНОЙ РАБОТЕ №6
Администрирование систем и сетей

по теме:

Создание WLAN

Желаемая оценка: 3

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Санкт-Петербург 2025

СОДЕРЖАНИЕ

ПЛАН РАБОТЫ

1. Настройка подключения к проводной сети.

2. Настройка точек доступа и перевод их в режим онлайн.

(1) Создание групп точек доступа и добавление точек доступа с одинаковой конфигурацией в одну группу для унифицированной настройки.

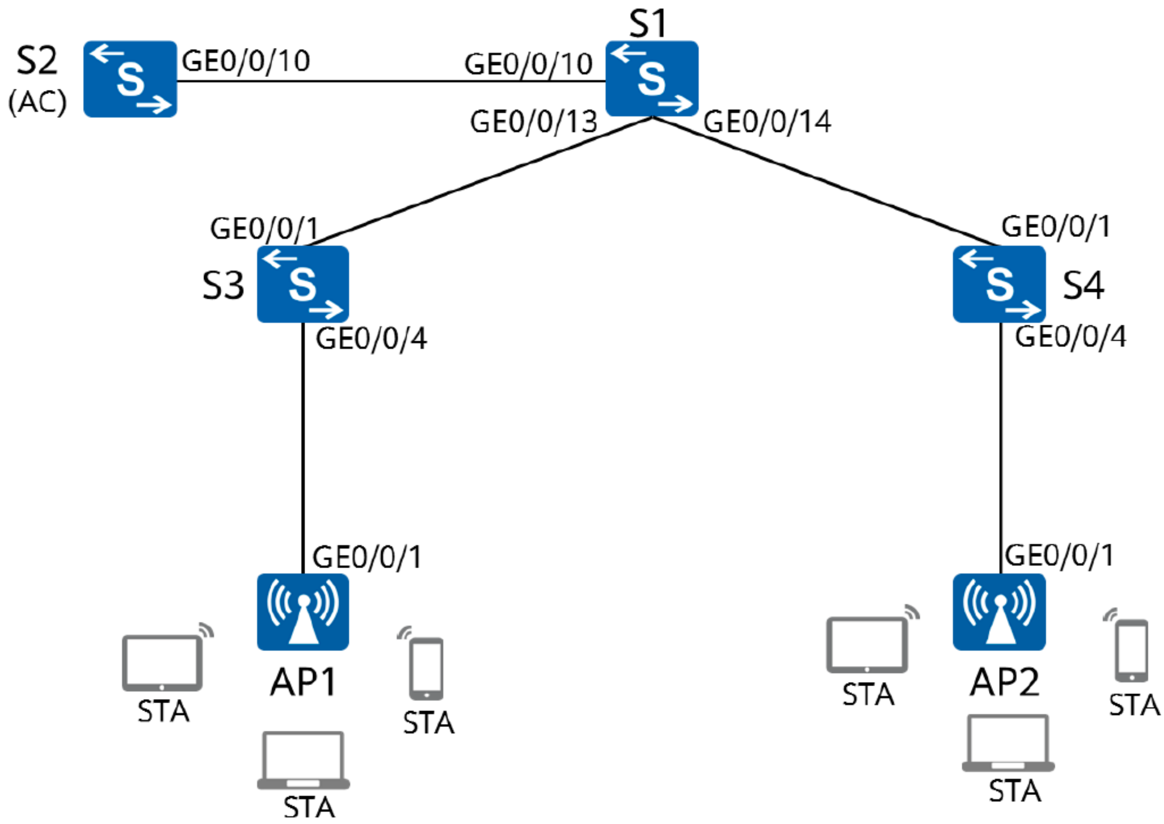
(2) Настройка системных параметров контроллера доступа, включая код страны и интерфейс-источник, используемый контроллером для связи с точками доступа.

(3) Настройка режима аутентификации AP и импорт AP для выхода точек доступа в сеть.

3. Настройка параметров сервисов WLAN и передача конфигурации точкам доступа, чтобы обеспечить доступ STA к WLAN.

1 Ход работы

1.1 Настройка основных параметров устройства



POE подключен автоматически

1.2 Настройка параметров проводной сети

Настройка VLAN

```
[S1]vlan batch 100 101
[S1]interface GigabitEthernet0/0/2
[S1-GigabitEthernet0/0/2]port link-type trunk
[S1-GigabitEthernet0/0/2]port trunk allow-pass vlan 100 101
[S1-GigabitEthernet0/0/2]quit
[S1]interface GigabitEthernet0/0/3
[S1-GigabitEthernet0/0/3]port link-type trunk
[S1-GigabitEthernet0/0/3]port trunk allow-pass vlan 100 101
[S1-GigabitEthernet0/0/3]quit
[S1]interface GigabitEthernet0/0/1
[S1-GigabitEthernet0/0/1]port link-type trunk
[S1-GigabitEthernet0/0/1]port trunk allow-pass vlan 100 101
```

```
[AC]vlan batch 100 101
Info: This operation may take a few seconds. Please wait for a moment...done.
[AC]interface GigabitEthernet 0/0/1
[AC-GigabitEthernet0/0/1]port link-type trunk
[AC-GigabitEthernet0/0/1]port trunk allow-pass vlan 100 101
[AC-GigabitEthernet0/0/1]quit
```

```
[S3]vlan batch 100 101
[S3]interface GigabitEthernet 0/0/2
[S3-GigabitEthernet0/0/2]port link-type trunk
[S3-GigabitEthernet0/0/2]port trunk allow-pass vlan 100 101
[S3-GigabitEthernet0/0/2]quit
[S3]interface GigabitEthernet 0/0/1
[S3-GigabitEthernet0/0/1]port link-type trunk
[S3-GigabitEthernet0/0/1]port trunk pvid vlan 100
[S3-GigabitEthernet0/0/1]port trunk allow-pass vlan 100 101
[S3-GigabitEthernet0/0/1]quit
```

```
[S4]interface GigabitEthernet0/0/3
[S4-GigabitEthernet0/0/3]port link-type trunk
[S4-GigabitEthernet0/0/3]port trunk allow-pass vlan 100 to 101
[S4-GigabitEthernet0/0/3]quit
[S4]interface GigabitEthernet0/0/1
[S4-GigabitEthernet0/0/1]port link-type trunk
[S4-GigabitEthernet0/0/1]port trunk pvid vlan 100
[S4-GigabitEthernet0/0/1]port trunk allow-pass vlan 100 to 101
[S4-GigabitEthernet0/0/1]quit
```

Настройка IP адресов интерфейсов

```
[S1]interface Vlanif 101
[S1-Vlanif101]ip address 192.168.101.254 24
[S1-Vlanif101]quit
[S1]interface LoopBack 0
[S1-LoopBack0]ip address 10.0.1.1 32
[S1-LoopBack0]interface Vlanif 100
[S1-Vlanif100]ip address 192.168.100.254 24
```

Настройка DHCP

```
[AC]interface Vlanif 100
[AC-Vlanif100]ip address 192.168.100.254 24
```

```
[S1]dhcp enable
[S1]ip pool sta
[S1-ip-pool-sta]network 192.168.101.0 mask 24
[S1-ip-pool-sta]gateway-list 192.168.101.254
[S1-ip-pool-sta]
[S1-ip-pool-sta]quit
```

```
[S1]interface Vlanif 101
[S1-Vlanif101]dhcp select global
[S1-Vlanif101]quit
```

```
[AC]dhcp enable
[AC]ip pool ap
Info: It is successful to create an IP address pool.
[AC-ip-pool-ap]network 192.168.100.254 mask 24
[AC-ip-pool-ap]gateway-list 192.168.100.254
[AC-ip-pool-ap]quit
[AC]interface Vlanif 100
[AC-Vlanif100]dhcp select global
[AC-Vlanif100]quit
```

1.3 Настройка параметров точек доступа для выхода в сеть

```
[AC]wlan
[AC-wlan-view]ap-group name ap-group1
[AC-wlan-ap-group-ap-group1]quit
```

Создаем профиль регулирующего домена и настраиваем код страны AC в профиле

```
[AC-wlan-view]regulatory-domain-profile name default
```

Настройка кода страны

```
[AC-wlan-regulate-domain-default]country-code cn
```

```
[AC]wlan
[AC-wlan-view]ap-group name ap-group1
[AC-wlan-ap-group-ap-group1]regulatory-domain-profile default
```

Настройка интерфейса

```
[AC-Vlanif100]capwap source interface Vlanif 100
```

Настройка аутентификации точек доступа

```
[AC-Vlanif100]capwap source interface Vlanif 100
```

Настройка аутентификации точек доступа

```
[AC]wlan
```

```
[AC-wlan-view]ap auth-mode mac-auth
```

Настройка MAC адресов для аутентификации

Узнаем MAC на AP1

```
[Huawei]display interface GigabitEthernet 0/0/1
```

```
00e0-fcc1-1570
```

Узнаем MAC на AP2

```
00e0-fcc1-1570
```

Добавляем их в AC

```
[AC-wlan-view]ap-id 0 ap-mac 00e0-fcc1-1570
[AC-wlan-ap-0]ap-id 1 ap-mac 00e0-fc71-3a80
```

Настройка группы AP в AC

```
[AC-wlan-view]ap-id 0
[AC-wlan-ap-0]ap-name ap1
[AC-wlan-ap-0]ap-group ap-group1
```

```
[AC-wlan-view]ap-id 0
[AC-wlan-ap-0]ap-name ap1
[AC-wlan-ap-0]ap-group ap-group1
```

Вывод информации о текущей WLAN на AC

```
[AC]wlan
[AC-wlan-view]display ap all
Info: This operation may take a few seconds. Please wait for a moment.done.
Total AP information:
nor  : normal          [2]
---------------------------------------------------------------------------
------
--------------
ID   MAC            Name Group     IP              Type            State STA
Upt
ime
---------------------------------------------------------------------------
------
--------------
0    00e0-fcc1-1570 ap1  ap-group1 192.168.100.207 AP6050DN        nor   0
3M:
```

```
23S
1    00e0-fc71-3a80 ap2  ap-group1 192.168.100.217 AP6050DN        nor   0
1M:
34S
----------------------------------------------------------------------------
------
--------------
Total: 2
```

1.4 Настройка параметров сервиса WLAN

```
[AC-wlan-view]security-profile name HCIA-WLAN
[AC-wlan-sec-prof-HCIA-WLAN]security wpa-wpa2 psk pass-phrase HCIA-Datacom
aes
[AC-wlan-sec-prof-HCIA-WLAN]quit
```

Создание профиля SSID HCIA-WLAN

```
[AC-wlan-view]ssid-profile name HCIA-WLAN
[AC-wlan-ssid-prof-HCIA-WLAN]ssid HCIA-WLAN
[AC-wlan-ssid-prof-HCIA-WLAN]quit
```

Создание профиля VAP HCIA-WLAN, применение профиля безопасности

```
[AC]wlan
[AC-wlan-view]vap-profile name HCIA-WLAN
```

Настройка режима передачи

```
[AC-wlan-vap-prof-HCIA-WLAN]forward-mode direct-forward
```

Настройка сервисной VLAN

```
[AC-wlan-vap-prof-HCIA-WLAN]service-vlan vlan-id 101
```

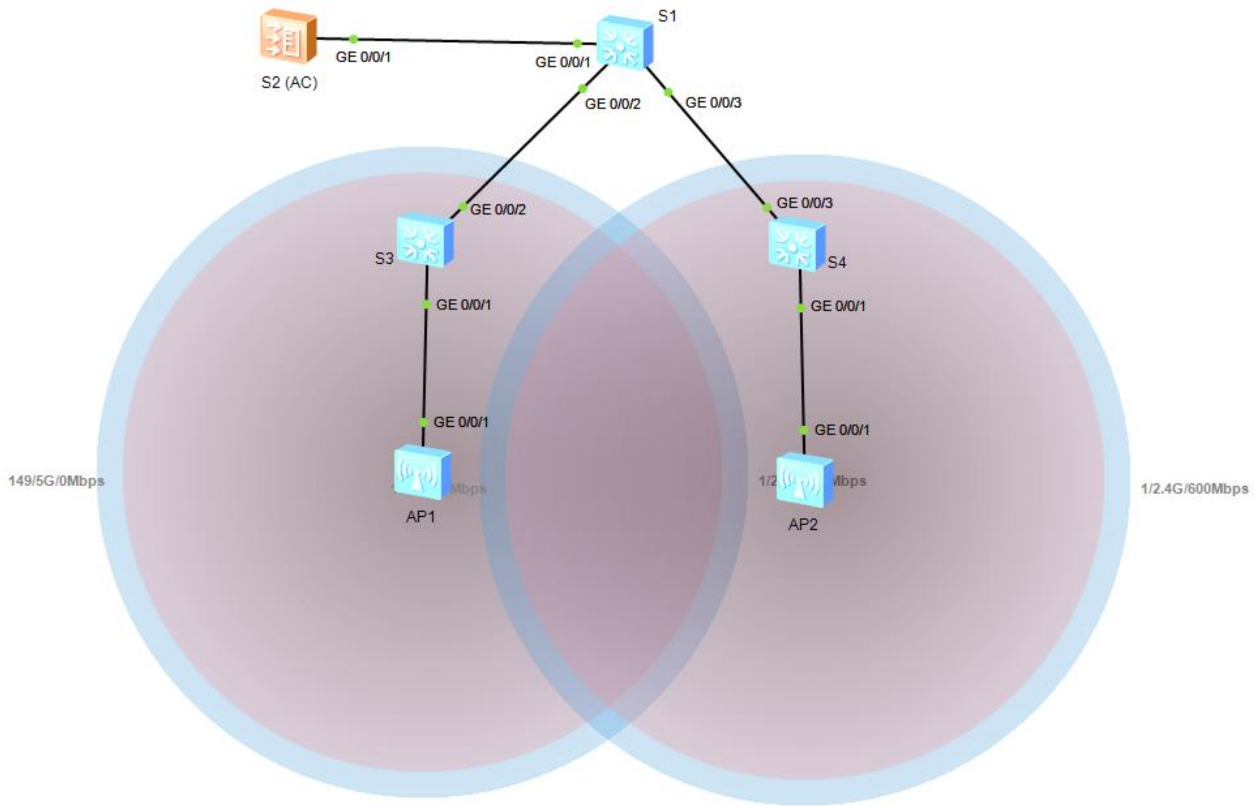
Применение профилей

```
[AC-wlan-vap-prof-HCIA-WLAN]security-profile HCIA-WLAN
[AC-wlan-vap-prof-HCIA-WLAN]ssid-profile HCIA-WLAN
[AC-wlan-vap-prof-HCIA-WLAN]quit
```

Привязка профиля AP к группе AP и примение конфигурации профиля VAP HCIA-WLAN к радиомодулю 0 и радиомодулю 1 точек доступа в группе AP

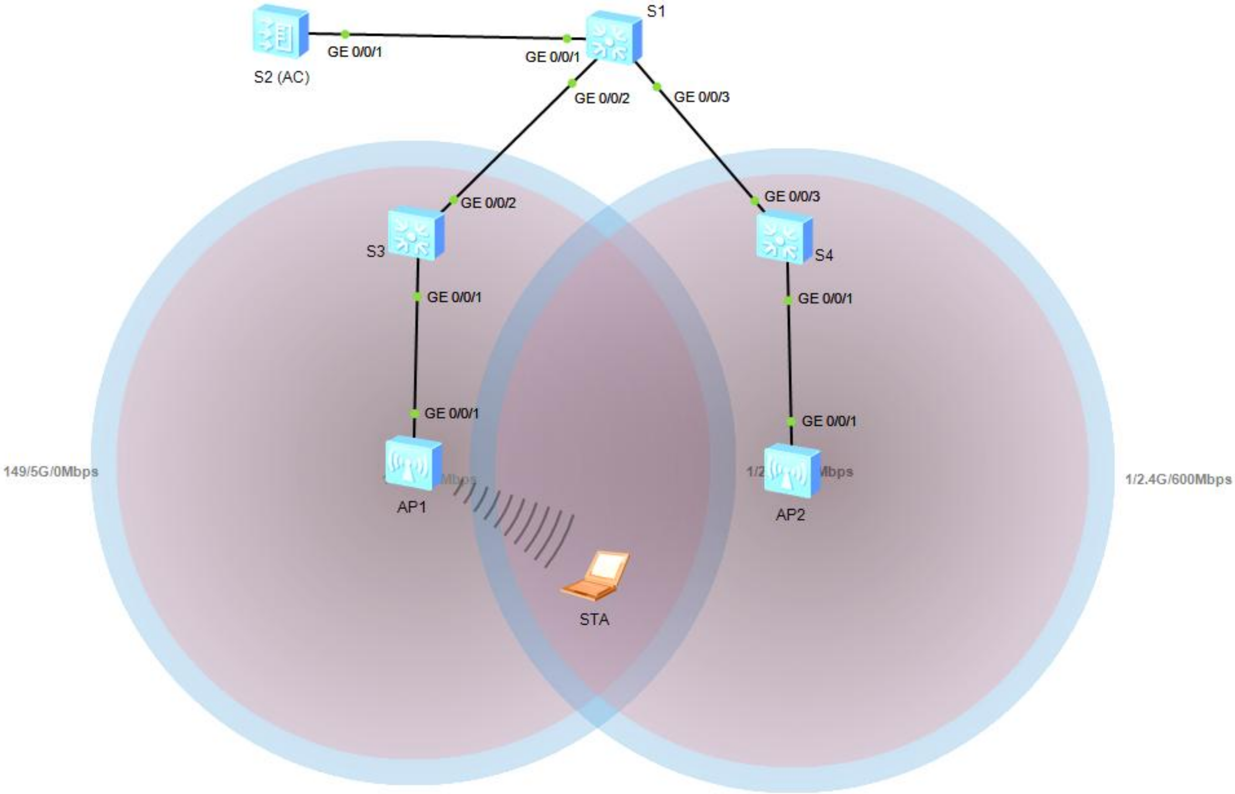
```
[AC-wlan-view]ap-group name ap-group1
```

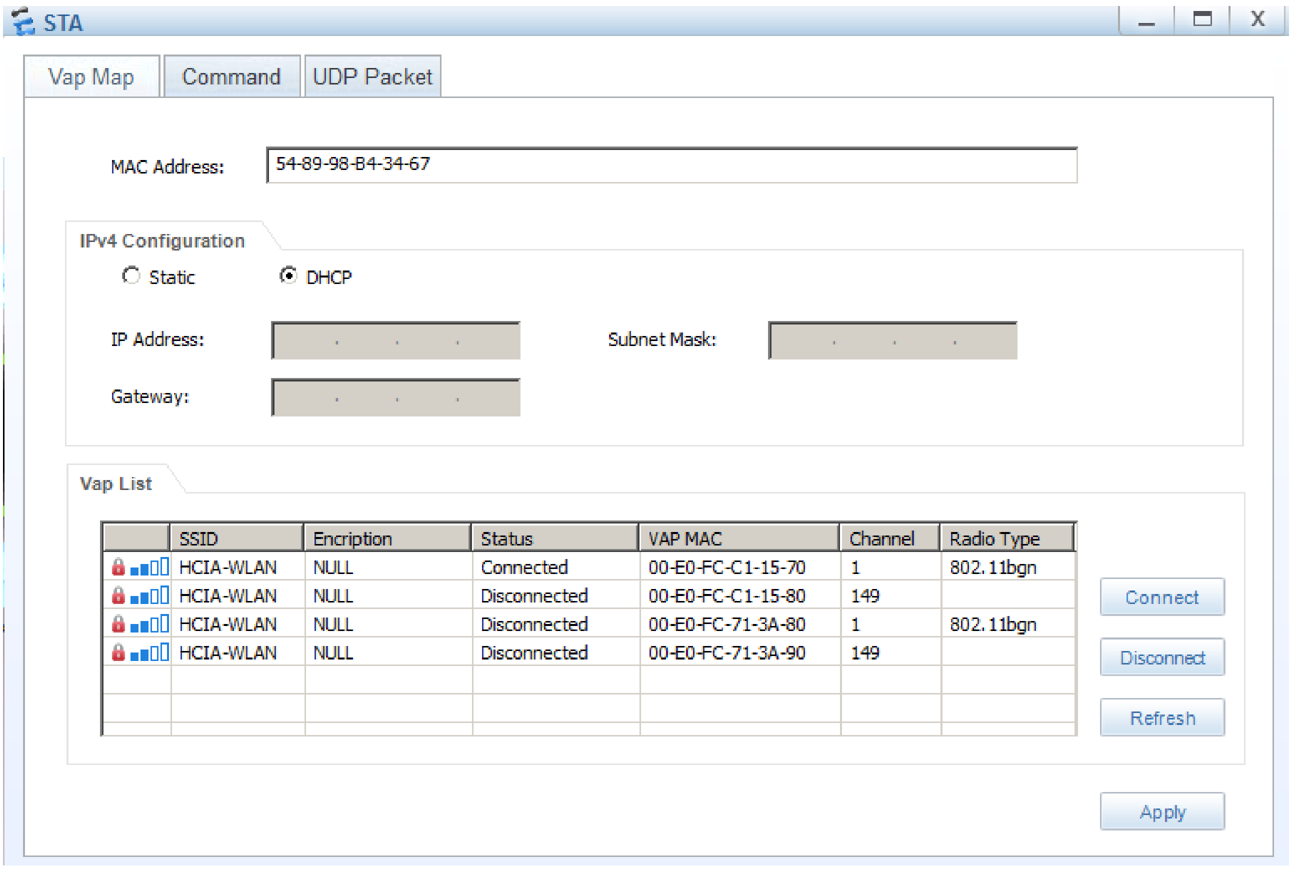
```
[AC-wlan-ap-group-ap-group1]vap-profile HCIA-WLAN wlan 1 radio all
```



Проверка

Подключение STA


S1
S2 (AC)
GE 0/0/1
GE 0/0/1
GE 0/0/2
GE 0/0/3
GE 0/0/2
GE 0/0/3
S3
S4
GE 0/0/1
GE 0/0/1
GE 0/0/1
GE 0/0/1
149/5G/0Mbps
1/2.4G/600Mbps
AP1
AP2
STA


STA
Vap Map
Command
UDP Packet
MAC Address:
54-89-98-B4-34-67
IPv4 Configuration
Static
DHCP
IP Address:
Subnet Mask:
Gateway:
Vap List
SSID | Encription | Status | VAP MAC | Channel | Radio Type
HCIA-WLAN | NULL | Connected | 00-E0-FC-C1-15-70 | 1 | 802.11bgn
HCIA-WLAN | NULL | Disconnected | 00-E0-FC-C1-15-80 | 149
HCIA-WLAN | NULL | Disconnected | 00-E0-FC-71-3A-80 | 1 | 802.11bgn
HCIA-WLAN | NULL | Disconnected | 00-E0-FC-71-3A-90 | 149
Connect
Disconnect
Refresh
Apply

Выданный IP

```
STA>ipconfig

Link local IPv6 address...........: ::
IPv6 address......................: :: / 128
IPv6 gateway......................: ::
IPv4 address......................: 192.168.101.253
Subnet mask.......................: 255.255.255.0
Gateway...........................: 192.168.101.254
Physical address..................: 54-89-98-B4-34-67
DNS server........................:

STA>
```

Ping S1 Loopback 0

```
STA>ping 10.0.1.1

Ping 10.0.1.1: 32 data bytes, Press Ctrl_C to break
From 10.0.1.1: bytes=32 seq=1 ttl=255 time=203 ms
From 10.0.1.1: bytes=32 seq=2 ttl=255 time=203 ms
From 10.0.1.1: bytes=32 seq=3 ttl=255 time=203 ms
From 10.0.1.1: bytes=32 seq=4 ttl=255 time=204 ms
From 10.0.1.1: bytes=32 seq=5 ttl=255 time=203 ms

--- 10.0.1.1 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 203/203/204 ms

STA>
```

Проверка информации об STA на AC

```
[AC]display station all
Rf/WLAN: Radio ID/WLAN ID
Rx/Tx: link receive rate/link transmit rate(Mbps)
---------------------------------------------------------------------------
------
----------------------
STA MAC          AP ID Ap name  Rf/WLAN  Band  Type  Rx/Tx      RSSI  VLAN
IP a
ddress      SSID
---------------------------------------------------------------------------
------
----------------------
5489-98b4-3467   0     ap1      0/1       2.4G  -     -/-         -     101
192.
168.101.253 HCIA-WLAN
```

```
------------------------------------------------------------------------------
------
----------------------
Total: 1 2.4G: 1 5G: 0
[AC]
```

Использованное оборудование

```
<S1>display current-configuration
#
sysname S1
#
vlan batch 100 to 101
#
cluster enable
ntdp enable
ndp enable
#
drop illegal-mac alarm
#
dhcp enable
#
diffserv domain default
#
drop-profile default
#
ip pool sta
 gateway-list 192.168.101.254
 network 192.168.101.0 mask 255.255.255.0
#
aaa
 authentication-scheme default
 authorization-scheme default
 accounting-scheme default
 domain default
 domain default_admin
 local-user admin password simple admin
 local-user admin service-type http
#
interface Vlanif1
#
interface Vlanif100
 ip address 192.168.100.254 255.255.255.0
#
interface Vlanif101
 ip address 192.168.101.254 255.255.255.0
 dhcp select global
#
interface MEth0/0/1
#
interface GigabitEthernet0/0/1
 port link-type trunk
 port trunk allow-pass vlan 100 to 101
```

```
#
interface GigabitEthernet0/0/2
 port link-type trunk
 port trunk allow-pass vlan 100 to 101
#
interface GigabitEthernet0/0/3
 port link-type trunk
 port trunk allow-pass vlan 100 to 101
#
interface GigabitEthernet0/0/4
#
interface GigabitEthernet0/0/5
#
interface GigabitEthernet0/0/6
#
interface GigabitEthernet0/0/7
#
interface GigabitEthernet0/0/8
#
interface GigabitEthernet0/0/9
#
interface GigabitEthernet0/0/10
#
interface GigabitEthernet0/0/11
#
interface GigabitEthernet0/0/12
#
interface GigabitEthernet0/0/13
#
interface GigabitEthernet0/0/14
#
interface GigabitEthernet0/0/15
#
interface GigabitEthernet0/0/16
#
interface GigabitEthernet0/0/17
#
interface GigabitEthernet0/0/18
#
interface GigabitEthernet0/0/19
#
interface GigabitEthernet0/0/20
#
interface GigabitEthernet0/0/21
#
interface GigabitEthernet0/0/22
#
interface GigabitEthernet0/0/23
#
interface GigabitEthernet0/0/24
#
interface NULL0
#
interface LoopBack0
```

```
 ip address 10.0.1.1 255.255.255.255
#
user-interface con 0
user-interface vty 0 4
#
return
```

```
<AC>display current-configuration
#
 sysname AC
#
 undo http secure-server enable
#
 set memory-usage threshold 0
#
ssl renegotiation-rate 1
#
vlan batch 100 to 101
#
authentication-profile name default_authen_profile
authentication-profile name dot1x_authen_profile
authentication-profile name mac_authen_profile
authentication-profile name portal_authen_profile
authentication-profile name macportal_authen_profile
#
dhcp enable
#
diffserv domain default
#
radius-server template default
#
pki realm default
 rsa local-key-pair default
 enrollment self-signed
#
ike proposal default
 encryption-algorithm aes-256
 dh group14
 authentication-algorithm sha2-256
 authentication-method pre-share
 integrity-algorithm hmac-sha2-256
 prf hmac-sha2-256
#
free-rule-template name default_free_rule
#
portal-access-profile name portal_access_profile
#
ip pool ap
 gateway-list 192.168.100.254
 network 192.168.100.0 mask 255.255.255.0
#
aaa
 authentication-scheme default
```

```
 authentication-scheme radius
  authentication-mode radius
 authorization-scheme default
 accounting-scheme default
 domain default
  authentication-scheme radius
  radius-server default
 domain default_admin
  authentication-scheme default
 local-user admin password irreversible-cipher
$1a$3uLL'kWye#$a^w|:yk,TLLbg'&uhy
}6!mBD6vw'4(p-hPRifI45$
 local-user admin service-type http
#
interface Vlanif100
 ip address 192.168.100.254 255.255.255.0
 dhcp select global
#
interface GigabitEthernet0/0/1
 port link-type trunk
 port trunk allow-pass vlan 100 to 101
#
interface GigabitEthernet0/0/2
#
interface GigabitEthernet0/0/3
#
interface GigabitEthernet0/0/4
#
interface GigabitEthernet0/0/5
#
interface GigabitEthernet0/0/6
#
interface GigabitEthernet0/0/7
 undo negotiation auto
 duplex half
#
interface GigabitEthernet0/0/8
 undo negotiation auto
 duplex half
#
interface NULL0
#
 snmp-agent local-engineid 800007DB03000000000000
 snmp-agent
#
ssh server secure-algorithms cipher aes256_ctr aes128_ctr
ssh server key-exchange dh_group14_sha1
ssh client secure-algorithms cipher aes256_ctr aes128_ctr
ssh client secure-algorithms hmac sha2_256
ssh client key-exchange dh_group14_sha1
#
capwap source interface vlanif100
#
user-interface con 0
```

```
user-interface vty 0 4
 protocol inbound all
user-interface vty 16 20
 protocol inbound all
#
wlan
 traffic-profile name default
 security-profile name default
 security-profile name HCIA-WLAN
  security              wpa-wpa2              psk                pass-phrase
%^%#l$I9VJzSt;.@/w;WacY2[Lj3PM`Nh~k[L\'FqoH;
%^%# aes
 security-profile name default-wds
 security-profile name default-mesh
 ssid-profile name default
 ssid-profile name HCIA-WLAN
  ssid HCIA-WLAN
 vap-profile name default
 vap-profile name HCIA-WLAN
  service-vlan vlan-id 101
  ssid-profile HCIA-WLAN
  security-profile HCIA-WLAN
 wds-profile name default
 mesh-handover-profile name default
 mesh-profile name default
 regulatory-domain-profile name default
 air-scan-profile name default
 rrm-profile name default
 radio-2g-profile name default
 radio-5g-profile name default
 wids-spoof-profile name default
 wids-profile name default
 wireless-access-specification
 ap-system-profile name default
 port-link-profile name default
 wired-port-profile name default
 serial-profile name preset-enjoyor-toeap
 ap-group name default
 ap-group name ap-group1
  radio 0
   vap-profile HCIA-WLAN wlan 1
  radio 1
   vap-profile HCIA-WLAN wlan 1
  radio 2
   vap-profile HCIA-WLAN wlan 1
 ap-id 0 type-id 56 ap-mac 00e0-fcc1-1570 ap-sn 21023544831076312476
  ap-name ap1
  ap-group ap-group1
 ap-id 1 type-id 56 ap-mac 00e0-fc71-3a80 ap-sn 21023544831062687B1F
  ap-name ap2
  ap-group ap-group1
 provision-ap
#
dot1x-access-profile name dot1x_access_profile
```

```
#
mac-access-profile name mac_access_profile
#
return
<AC>
```

```
<S3>display current-configuration
#
sysname S3
#
vlan batch 100 to 101
#
cluster enable
ntdp enable
ndp enable
#
drop illegal-mac alarm
#
diffserv domain default
#
drop-profile default
#
aaa
 authentication-scheme default
 authorization-scheme default
 accounting-scheme default
 domain default
 domain default_admin
 local-user admin password simple admin
 local-user admin service-type http
#
interface Vlanif1
#
interface MEth0/0/1
#
interface GigabitEthernet0/0/1
 port link-type trunk
 port trunk pvid vlan 100
 port trunk allow-pass vlan 100 to 101
#
interface GigabitEthernet0/0/2
 port link-type trunk
 port trunk allow-pass vlan 100 to 101
#
interface GigabitEthernet0/0/3
#
interface GigabitEthernet0/0/4
#
interface GigabitEthernet0/0/5
#
interface GigabitEthernet0/0/6
#
interface GigabitEthernet0/0/7
```

```
#
interface GigabitEthernet0/0/8
#
interface GigabitEthernet0/0/9
#
interface GigabitEthernet0/0/10
#
interface GigabitEthernet0/0/11
#
interface GigabitEthernet0/0/12
#
interface GigabitEthernet0/0/13
#
interface GigabitEthernet0/0/14
#
interface GigabitEthernet0/0/15
#
interface GigabitEthernet0/0/16
#
interface GigabitEthernet0/0/17
#
interface GigabitEthernet0/0/18
#
interface GigabitEthernet0/0/19
#
interface GigabitEthernet0/0/20
#
interface GigabitEthernet0/0/21
#
interface GigabitEthernet0/0/22
#
interface GigabitEthernet0/0/23
#
interface GigabitEthernet0/0/24
#
interface NULL0
#
user-interface con 0
user-interface vty 0 4
#
return
```

```
<S4>display current-configuration
#
sysname S4
#
vlan batch 100 to 101
#
cluster enable
ntdp enable
ndp enable
#
drop illegal-mac alarm
```

```
#
diffserv domain default
#
drop-profile default
#
aaa
 authentication-scheme default
 authorization-scheme default
 accounting-scheme default
 domain default
 domain default_admin
 local-user admin password simple admin
 local-user admin service-type http
#
interface Vlanif1
#
interface MEth0/0/1
#
interface GigabitEthernet0/0/1
 port link-type trunk
 port trunk pvid vlan 100
 port trunk allow-pass vlan 100 to 101
#
interface GigabitEthernet0/0/2
#
interface GigabitEthernet0/0/3
 port link-type trunk
 port trunk allow-pass vlan 100 to 101
#
interface GigabitEthernet0/0/4
#
interface GigabitEthernet0/0/5
#
interface GigabitEthernet0/0/6
#
interface GigabitEthernet0/0/7
#
interface GigabitEthernet0/0/8
#
interface GigabitEthernet0/0/9
#
interface GigabitEthernet0/0/10
#
interface GigabitEthernet0/0/11
#
interface GigabitEthernet0/0/12
#
interface GigabitEthernet0/0/13
#
interface GigabitEthernet0/0/14
#
interface GigabitEthernet0/0/15
#
interface GigabitEthernet0/0/16
```

```
#
interface GigabitEthernet0/0/17
#
interface GigabitEthernet0/0/18
#
interface GigabitEthernet0/0/19
#
interface GigabitEthernet0/0/20
#
interface GigabitEthernet0/0/21
#
interface GigabitEthernet0/0/22
#
interface GigabitEthernet0/0/23
#
interface GigabitEthernet0/0/24
#
interface NULL0
#
user-interface con 0
user-interface vty 0 4
#
return
```

ЗАКЛЮЧЕНИЕ

Цель работы достигнута.